Run and Manage Docker Containers

Both Dockerfile and Image is created:

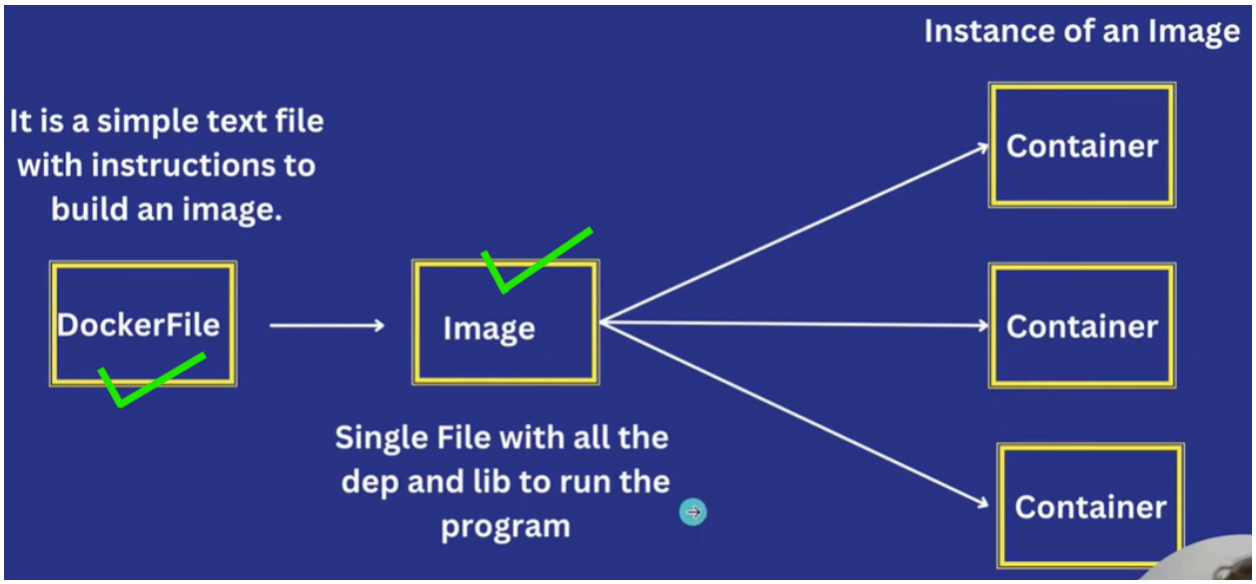
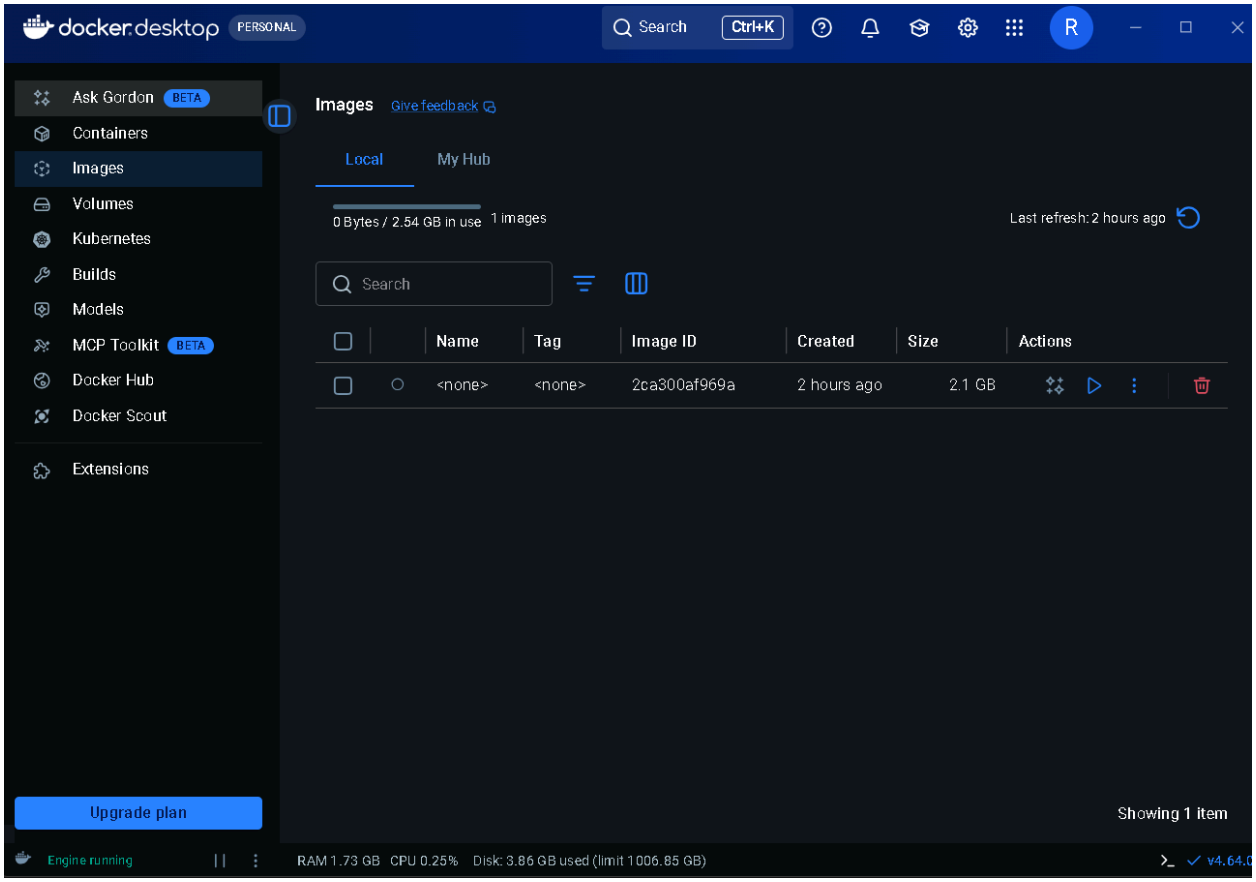


Image is like you already have the application, and just need to run the application. In docker file, its mentioned, during creation of image, npm install to bring back the node_modules,
If you face error like:

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application> cd testapp
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker image ls -a
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngir
The system cannot find the file specified.
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp>
```

Open the docker Desktop:



Then run the command again:

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker image ls -a

IMAGE          ID             DISK USAGE   CONTENT SIZE   EXTRA
<untagged>     2ca300af969a       2.11GB          491MB
```

In docker desktop:

Name	Tag	Image ID	Created	Size	Actions
<none>	<none>	2ca300af969a	2 hours ago	2.1 GB	

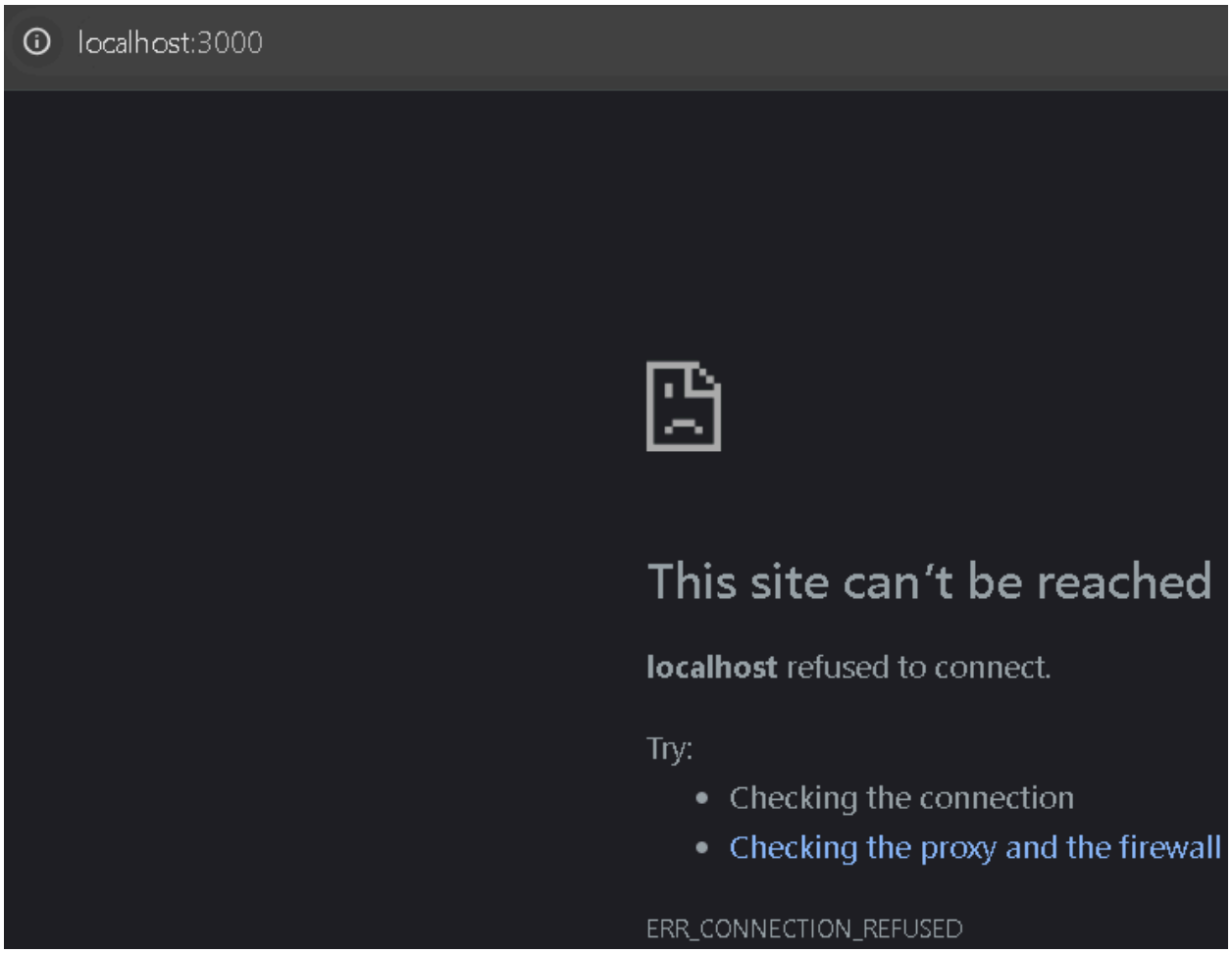
We can create multiple container using the same image:
You can run the application/image multiple times.
To run the image: **docker run 2ca300af969a**

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker run 2ca300af969a

> testapp@0.1.0 start
> react-scripts start

(node:26) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'on
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:26) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: '
Starting the development server...

Compiled successfully!
```



Above the machine/operating system, there exists the container which contains the application, so laptop's local machine browser can't able to access the application that is within the container or isolated environment.

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS      NAMES
a4c459af29e6   2ca300af969a   "docker-entrypoint.s…"   6 minutes ago   Up 6 minutes   3000/tcp   naughty_gould
```

Name	Container ID	Image	Port(s)
naughty_gould	a4c459af29e6	2ca300af9	

To stop the container:

```
\testapp> docker ps
     CREATED          STATUS          PORTS       NAMES
…"   6 minutes ago    Up 6 minutes    3000/tcp    naughty_gould
\testapp> docker stop naughty_gould

\testapp> docker ps
S    PORTS      NAMES
\testapp>
```

Inside the container, port 3000, web app is accessible, which is not outside container,

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker image ls -a

IMAGE          ID             DISK USAGE   CONTENT SIZE   EXTRA
<untagged>     2ca300af969a       2.11GB          491MB    U
```

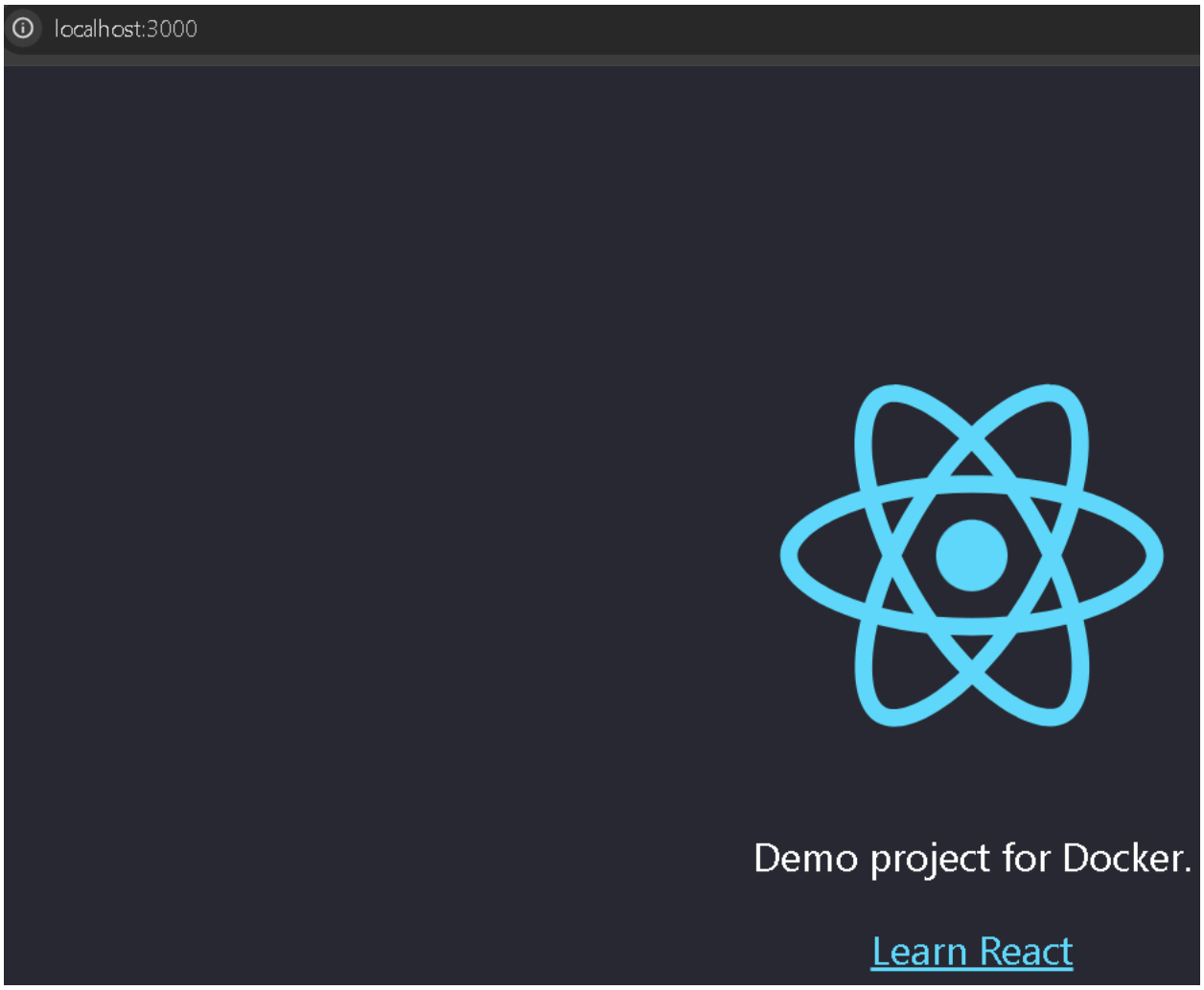
-p for port binding. Inside container, which has port 3000 for listening to web application. To access the web application outside the container need to bind 3000 to the inner 3000. In short, we are making access outside the container. Keeping the same port.

```
\testapp> docker run -p 3000:3000 2ca300af969a
```

```
You can now view testapp in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://172.17.0.2:3000
```

Now reload the page:



So finally we can access the project.